Statistics and R cheat sheet

Concepts of Biostatistics and Data Science: a two-day crash course with practical analysis in R

All examples use the course dataset `hearts` (the Healthy Hearts District Study), loaded with `hearts <- readRDS("data/hearts.rds")`. The data are **simulated for teaching**: the results illustrate methods and say nothing about real people. Packages: `readr`, `dplyr`, `ggplot2`, `broom`; `MASS` and `cluster` come with R. For choosing a method, use the decision diagrams D0 to D6.

Before any analysis: five questions

1. **Question.** Describe, compare groups, assess an association, predict, or discover groups?
2. **Outcome type.** Continuous, binary, nominal, ordinal, count or time-to-event?
3. **Design.** Independent, paired, repeated or clustered observations?
4. **Groups or predictors.** Two groups, more than two, one predictor, several?
5. **Effect to report.** Difference, ratio or slope, with its 95% CI. A p-value alone is not an answer.

Reading results honestly

- **Statistical significance is not practical importance.** A tiny difference can be "significant" in a large study. Judge size from the estimate and CI.
- **Significance is not causation.** In observational data, confounding, bias and chance can all produce associations.
- **"Not significant" is not "no effect".** Look at the CI: it may include both no effect and important effects.
- A p-value is not the probability that the null hypothesis is true.

Glossary

Observation one unit measured (here, one participant; one row).

Variable a characteristic recorded for each observation (one column).

Population, sample the people we want to learn about; the subset we measured.

Estimate a number computed from the sample to describe the population (mean difference, OR, slope).

Standard error (SE) how much an estimate would vary from sample to sample.

95% confidence interval (CI) a range of population values compatible with the data. Across many studies, 95% of such intervals would contain the true value.

Null hypothesis a reference claim, usually "no difference" or "no association".

p-value how surprising data at least this extreme would be if the null hypothesis were true.

Effect size the size of a difference or association, in units people understand.

Independent observations no observation tells you anything about another (no pairing, repeats or clusters).

Paired two measurements linked to the same person or a matched pair.

Residual observed value minus the value fitted by a model.

Rank position of a value when all values are sorted (1 = smallest).

Skewed a distribution with a long tail on one side (CRP has a long right tail).

Confounder a common cause of the exposure and the outcome. Adjust for it.

Mediator a variable on the path from exposure to outcome. Do not adjust for it if you want the total effect.

Effect modification the association differs between subgroups (e.g. women and men).

Odds number with the outcome divided by number without it.

Offset a term that tells a count model how long each person was observed.

Overdispersion counts vary more than a Poisson model assumes.

Censoring the event has not (yet) been observed when follow-up ends.

Scaling (standardising) subtract the mean and divide by the SD, so variables share a scale.

Describing data

Variable	Summary	Plot
Binary, nominal	n and % (state the denominator)	bar chart
Ordinal	n and % in order; median (IQR)	ordered bars
Continuous, symmetric	mean (SD)	histogram, boxplot
Continuous, skewed	median (IQR)	histogram (log scale)
Count	mean, variance; rate per person-time	bar chart
Two categorical	table with row %	bars of proportions
Continuous by group	summary per group	boxplot with points
Two continuous	correlation	scatterplot

```
count(hearts, district) |> mutate(pct = 100 * n / sum(n))
hearts |> group_by(sex) |> summarise(n = n(),
  mean = mean(sbp_0), sd = sd(sbp_0))
median(hearts$crp, na.rm = TRUE)
ggplot(hearts, aes(x = district, y = sbp_0)) +
  geom_boxplot() + geom_jitter(width = 0.15)
```

Course data: SBP mean 129.3 (SD 15.6) mmHg, n = 480. CRP is right-skewed: median 1.9 (IQR 1.2 to 3.7) mg/L, n = 466 (14 missing); its mean (3.0) is pulled up by a few high values.

Comparing a continuous or ordinal outcome

Welch t-test: two independent groups

Question Do two groups differ in their mean outcome?

Data Continuous outcome; two independent groups.

Assumptions Independent observations; group means approximately normal (usually fine in moderate or large groups unless there is strong skew or outliers). Welch does not assume equal variances, so it is a sensible default. Check: boxplots or histograms by group.

R
```
t.test(sbp_0 ~ sex, data = hearts)
```

Report Difference in means with 95% CI.

Course Women 126.4, men 132.5 mmHg. Difference (women minus men) −6.1 mmHg (95% CI −8.9 to −3.4).

In words On average women's SBP was about 6 mmHg lower; the data are compatible with a difference of about 3 to 9 mmHg.

Mistake Ignoring which group is subtracted (R uses the factor level order), or reporting only the p-value.

Paired t-test: the same people measured twice

Question Did the mean change within people (or within matched pairs)?

Data Continuous outcome measured twice on the same unit.

Assumptions Pairs independent of each other; differences approximately normal or enough pairs. Check: histogram of the differences.

R
```
t.test(prog$sbp_12, prog$sbp_0, paired = TRUE)
```

Report Mean difference with 95% CI.

Course Lifestyle group (n = 224): mean change −6.0 mmHg (95% CI −7.2 to −4.8).

Mistake Analysing paired data as independent groups (CI widens to −9.1 to −2.9). And a within-group change is not the programme effect: compare the change between the randomised groups (4.0 mmHg greater fall, 95% CI 2.3 to 5.6).

Wilcoxon rank-sum (Mann-Whitney) test

Question Do values tend to be higher in one group than in the other?

Data Ordinal or continuous outcome; two independent groups.

Assumptions Independent observations. Reading it as a shift in location needs distributions of similar shape. It is **not assumption-free** and **not in general a test of medians**.

R
```
wilcox.test(crp ~ smoker, data = hearts,
            conf.int = TRUE)
```

Report Medians (IQR) per group, and the Hodges-Lehmann shift with CI (the median of all smoker versus non-smoker differences).

Course CRP median 1.8 (IQR 1.1 to 3.3) in non-smokers, 2.8 (1.7 to 5.3) in smokers; shift 0.9 mg/L (95% CI 0.5 to 1.4), $p < 0.001$. Alternative for a positive skewed outcome: t-test on `log(crp)` gives a ratio of geometric means of 1.63 (1.31 to 2.03).

Mistake Choosing it because "the sample is small" or because a normality test gave $p < 0.05$. Choose by the question and the data.

Wilcoxon signed-rank test: paired differences

Question Do the within-pair differences tend to be above or below zero?

Assumptions Independent pairs; for the pseudomedian estimate, differences roughly symmetric.

R
```
wilcox.test(prog$sbp_12, prog$sbp_0,
            paired = TRUE, conf.int = TRUE)
```

Course Pseudomedian change −6.0 mmHg (95% CI −7.0 to −5.0).

Mistake Using the rank-sum test (independent groups) on paired data.

One-way ANOVA and Welch's ANOVA: three or more groups

Question Do mean outcomes differ across three or more independent groups?

Assumptions Independent observations; approximately normal residuals; classical ANOVA also assumes equal variances, Welch's does not. Check: `plot(fit, which = 1:2)`.

R
```
oneway.test(sbp_0 ~ district, data = hearts)
fit <- aov(sbp_0 ~ district, data = hearts)
TukeyHSD(fit)
```

Report Group means with CIs; pairwise differences with adjusted CIs if needed.

Course Means: North 126, South 134, East 131, West 126 mmHg. Welch $F = 6.83$, $p < 0.001$. Tukey: South minus North 7.6 mmHg (2.5 to 12.8).

Mistake Concluding from the overall p-value **which** groups differ, or running many unadjusted t-tests.

Kruskal-Wallis test

Question Do values tend to differ across three or more independent groups (rank-based)?

Assumptions Independent observations. Not a test of medians in general.

R `kruskal.test(crp ~ district, data = hearts)`

Course Median CRP 1.8 to 2.5 mg/L across districts; p = 0.067. Weak evidence of a difference, which is not evidence of no difference.

Mistake Following up with unadjusted pairwise tests.

Categorical outcomes

Chi-square test of association

Question Is a categorical outcome associated with a categorical exposure (do the proportions differ)?

Data Two categorical variables; independent observations.

Assumptions Expected counts about 5 or more in each cell. Check: `chisq.test(tab)$expected`.

R
```
tab <- table(hearts$smoker, hearts$high_bp_0)
prop.table(tab, margin = 1)
chisq.test(tab)
```

Report Proportions in each group, and a difference or ratio with 95% CI (see the measures section).

Course High BP in 48% of smokers and 21% of non-smokers; χ^2 = 23.1, 1 df, p < 0.001.

Mistake Reporting only the p-value; using column percentages when the question needs row percentages; using it on paired data.

Fisher's exact test

Question As for chi-square, when expected counts are small.

R `fisher.test(tab_ae)`

Course Adverse events 14/240 with the programme vs 3/240 with standard advice; p = 0.011; RR 4.67 (95% CI 1.36 to 16.03). The wide CI shows how uncertain the size of the increase is.

Mistake Thinking a small total sample decides it. The expected counts matter.

McNemar's test: paired binary data

Question Did the proportion with an outcome change within the same people?

R
```
tab_paired <- table(prog$high_bp_0,
prog$high_bp_12)
mcnemar.test(tab_paired)
```

Course Lifestyle group: 28 people moved from high BP to not high, 6 the other way; p < 0.001. Only these discordant pairs carry information about change.

Mistake A chi-square test on before and after as if they were different people.

Proportions, risks, odds and their ratios

Proportion = number with the characteristic / number in the group. Always say what the denominator is.

Risk = number who *develop* the outcome during a period / number at risk at the start. Needs follow-up (cohort study or trial).

Prevalence = number who *have* the outcome at one time / number observed then (cross-sectional study).

Odds = number with the outcome / number without it. A risk of 0.20 is odds of 0.20/0.80 = 0.25.

Risk difference (RD) = risk in exposed − risk in unexposed. Absolute; report in percentage points.

Risk ratio (RR) = risk in exposed / risk in unexposed. Relative. The same for prevalences gives a **prevalence ratio (PR)**.

Odds ratio (OR) = odds in exposed / odds in unexposed. From logistic regression with several predictors it is an **adjusted OR (aOR)**.

A value of 1 for a ratio, or 0 for a difference, means no association.

Cross-sectional example (baseline): smoking and high BP

	High BP	No high BP	Total
Smokers	38	41	79
Non-smokers	86	315	401

Prevalence: 38/79 = 0.481 vs 86/401 = 0.214

Prevalence difference: 0.267 (26.7 percentage points)

PR = 0.481 / 0.214 = **2.24** (95% CI 1.67 to 3.02)

Odds: 38/41 = 0.927 vs 86/315 = 0.273

OR = 0.927 / 0.273 = **3.39** (95% CI 2.06 to 5.61)

Adjusted for age and sex (logistic regression): aOR **2.04** (1.17 to 3.54)

Randomised trial example: adverse events

	Event	No event	Total
Lifestyle programme	14	226	240
Standard advice	3	237	240

Risk: 14/240 = 5.8% vs 3/240 = 1.25%; RD 4.6 percentage points

RR = **4.67** (95% CI 1.36 to 16.03); OR = 4.89 (1.39 to 17.26)

The outcome is rare, so OR and RR are close. For high BP (21% to 48%) the OR (3.39) is much further from 1 than the PR (2.24).

Which measures the design allows

Cohort, trial	Risks, RD, RR (and OR)
Cross-sectional	Prevalences, prevalence difference, PR (and OR)
Case-control	**Only the OR.** The proportion of cases is fixed by the design, so risks cannot be estimated.

Common mistakes

- Calling an OR of 3.4 “3.4 times the risk”. That is only roughly true for rare outcomes.
- Reporting a relative measure without the absolute risks behind it.
- Mixing up percent and percentage points.

Confounding, mediation and effect modification

Confounder: a common cause of exposure and outcome. Older people in the course data are less active and have higher SBP, so age confounds the activity and SBP association. *Adjust* for confounders.

Mediator: lies on the path from exposure to outcome (activity lowers BMI, BMI raises SBP). Adjusting for it removes part of the effect you want. Do not adjust for mediators when estimating a total effect.

Effect modification: the association differs between subgroups. Report the subgroup estimates. In the course data the smoking OR was 2.75 in women and 3.28 in men: similar, so a single adjusted estimate is reasonable.

Choose adjustment variables from subject knowledge (draw a diagram of the causes), not from p-values or stepwise selection.

SBP per extra hour/week of activity (mmHg)	
Crude	−2.08 (−2.62 to −1.53)
Adjusted for age, sex (confounders)	−1.60 (−2.08 to −1.12)
Also adjusted for BMI (mediator)	−0.86 (−1.36 to −0.37)
High BP, smokers vs non-smokers	
Crude OR	3.39 (2.06 to 5.61)
OR adjusted for sex (Mantel-Haenszel)	3.09 (1.85 to 5.15)
aOR adjusted for age and sex	2.04 (1.17 to 3.54)

The change from crude to adjusted shows the confounding by age and sex. Adjustment can only remove confounding by variables that were measured, and measured well.

Correlation and regression

Pearson and Spearman correlation

Question How strongly do two variables move together?

Data Two continuous (Spearman: also ordinal) variables; independent observations.

Assumptions Pearson: a linear pattern; sensitive to outliers. Spearman uses ranks and measures a monotonic pattern. Always look at the scatterplot.

R
```
cor.test(~ sbp_0 + age, data = hearts)
cor.test(~ crp + activity_hr, data = hearts,
         method = "spearman", exact = FALSE)
```

Course Age and SBP: r = 0.50 (95% CI 0.43 to 0.57). CRP and activity: Spearman rho = −0.15.

Mistake Reading r as the size of the change (use regression), or r near 0 as “no relationship” (it could be curved).

Simple linear regression

Question How much does the mean outcome change per unit of x?

Assumptions Independence (from the design); linearity; constant spread of residuals; approximately normal residuals (matters least in large samples). The outcome itself need not be normal. Check: `plot(fit)`.

R
```
fit <- lm(sbp_0 ~ age10, data = hearts)
tidy(fit, conf.int = TRUE)
```

Report Slope with 95% CI; R^2 if useful.

Course Mean SBP 6.6 mmHg higher per 10 years of age (95% CI 5.6 to 7.6); R^2 = 0.25.

Mistake Interpreting the intercept at age 0, or predicting outside the range of the data.

Multiple linear regression

Question What is the association of x with the outcome among people with the same values of the other predictors?

R
```
fit <- lm(sbp_0 ~ activity_hr + age + sex,
          data = hearts)
```

Report Adjusted coefficients with 95% CIs, and n used.

Course −1.60 mmHg per extra hour/week (95% CI −2.08 to −1.12), adjusted for age and sex (n = 473).

In words Among people of the same age and sex, each extra hour of weekly activity was associated with SBP about 1.6 mmHg lower on average.

Mistake Adjusting for mediators; choosing variables by p-value; not noticing that missing values change n between models.

Logistic regression: binary outcome

Question How are the odds of a binary outcome related to one or more predictors?

Assumptions Independent observations; linear on the log-odds scale for continuous predictors; enough events for the number of coefficients.

R
```
fit <- glm(high_bp_0 ~ smoker + age10 + sex,
           family = binomial, data = hearts)
tidy(fit, conf.int = TRUE, exponentiate = TRUE)
```

Report (Adjusted) odds ratios with 95% CIs; predicted probabilities help explain them.

Course Smoking aOR 2.04 (1.17 to 3.54); per 10 years of age 2.29 (1.84 to 2.87); men vs women 2.06 (1.30 to 3.30).

Mistake Reading an OR as a risk ratio when the outcome is common. R models the probability of the second factor level (here "Yes").

Poisson regression: counts and rates

Question How are event rates related to predictors?

Data Count outcome; follow-up time can differ between people.

Assumptions Independent observations; variance equal to the mean. Check overdispersion: Pearson χ^2 / residual df (about 1 is fine).

R
```
fit <- glm(clinic_visits ~ programme + age10 +
  high_bp_0 + offset(log(followup_months)),
  family = poisson, data = hearts)
```

Report Incidence rate ratios (IRR) with 95% CIs.

Course Dispersion 2.03, so Poisson CIs are too narrow. Programme IRR 0.91: Poisson CI 0.82 to 1.02, quasi-Poisson CI 0.78 to 1.06.

Mistake Comparing raw counts when follow-up differs (no offset), or ignoring overdispersion. Remedies: `family = quasipoisson` or `MASS::glm.nb()`.

Clustering (unsupervised learning)

k-means clustering

Question Are there groups of people with similar values on chosen variables? There is no outcome.

Data Numeric variables, complete cases. Scale variables measured in different units.

Choices Variables; scaling; k (elbow plot, average silhouette width, interpretability); several random starts.

R
```
z <- scale(life[, c("activity_min",
       "diet_score", "bmi")])
set.seed(2024)
km <- kmeans(z, centers = 3, nstart = 25)
```

Report How variables and k were chosen; cluster sizes; profiles on the original scale.

Course Clusters of 146, 228 and 89 people (low, middle and high activity and diet scores, with BMI falling across them). Average silhouette 0.29 for k = 3 and 0.34 for k = 2: weak structure.

Mistake Not scaling (activity minutes then dominate); treating clusters as real types; testing whether clusters differ on the variables used to build them.

Hierarchical clustering

Question As for k-means, shown as a tree (dendrogram) that can be cut at any k.

R
```
hc <- hclust(dist(z), method = "ward.D2")
plot(hc); cutree(hc, k = 3)
```

Choices Distance and linkage. Ward's method gives compact groups; average linkage gave groups of 426, 30 and 7 in the course data.

Mistake Reading the dendrogram as proof that groups exist.

R essentials

`library(dplyr)` load a package (install once with `install.packages("dplyr")`)

`raw <- read_csv("data/hearts_raw.csv")` import

`glimpse(raw); summary(raw); colSums(is.na(raw))`

`mutate(activity_min = na_if(activity_min, -99))` recode a missing-value code

`mutate(district = factor(district, levels = c(...)))` first level = reference group

`filter(hearts, programme == "Lifestyle programme")`

`select(hearts, id, sbp_0); count(hearts, smoker)`

`group_by(sex) |> summarise(mean = mean(bmi, na.rm = TRUE))`

`ggplot(data, aes(x = ..., y = ...)) + geom_point()`

`ggsave("outputs/plot.png", width = 6, height = 4)`

`y ~ x` a formula: outcome on the left, predictors on the right

`na.rm = TRUE` ignore missing values in a summary

`?t.test` open the help page for a function



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